

COMP2K

Quantum Computing

Laboratory – DRAFT

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Instructions

Recommended you complete this part by the end of [Week 12](#).
You should demo this lab in your [Week 13](#) practical session.

[You must demonstrate it to the instructor in one of your practical sessions BEFORE the due date in order to be awarded marks. Please check the ECP for the correct due date. Note that sections are 'complete' and marks are awarded by attempting each task AND correctly answering related questions to the satisfaction of the instructor.]

[Quantum computing](#) is a form of computation that uses quantum phenomena such as superposition and entanglement that forms an essential part of [quantum mechanics](#). Quantum mechanics describes physics of matter at the extraordinarily small scale surprisingly accurately and is the most successful physical theory of the universe we currently have being able to predict outcomes to an accuracy of up to 10⁻¹¹ decimal places! See for example Chapter 15 of (Moore and Mertens, 2011) for an introduction. Quantum computing's pervasiveness will only increase as it comes out of infancy and there continues to be major advancements as now there are known to be several quantum computers in existence. It is expected to become prominent part in computing and algorithms of the future by creating a new series of [quantum algorithms](#) that surpass the computational power of classical supercomputers by "harnessing" the power of quantum computation and is one of the 21st century's biggest challenges.

Quantum computers uses quantum bits or '[qubits](#)', bits that are in superposition of between states instead of being either on or off and being in one or the other state (like a classical binary bit). Qubits still collapse into binary bits, but its actual final state has a probabilistic outcome given by a probability density as opposed to a deterministic one.

In this laboratory we'll be using [Qiskit](#), an open-source framework for quantum computing that allows you to simulate quantum bits on classical computers. You can install Qiskit as a Python library through [here](#).

This laboratory will allow you to explore qubits and the different classical and quantum operations you can use to manipulate qubits in a quantum circuit. Qiskit's documentation can be found [here](#).

Section I – Microsoft Seminar

Microsoft released an extremely useful, self-contained seminar on quantum computing that is a very valuable resource if you find the concepts in this laboratory difficult to follow:

Seminar on Quantum Computing for Computer Scientists – https://youtu.be/F_RiqjdH2oM

Recently in 2025, 3Blue1Brown released a helpful video explaining the concepts required - https://youtu.be/RQWpF2Gb-gU?si=3rxnO1qd_ulx7NgJ

Try getting a handle of quantum computing concepts from the video and playing with the mathematics of the quantum mechanical operators BEFORE you start the following section. See for example equations (6.23) to (6.46) from Shakes' book. Chapter 15 of (Moore and Mertens, 2011) also provides a great introduction to the area, including all the necessary quantum mechanical preliminaries. Although you do not receive marks for this section and you do not have to complete the entire video, there is significant overlap with the video and the requirements of the next section that do award marks.

Section II – Qiskit (20 Marks)

Complete the following exercises using your knowledge of quantum computing, quantum algorithms and quantum circuits using Qiskit.

[See the relevant sections of Shakes' book, Chapter 15 of (Moore and Mertens, 2011) and the Microsoft Seminar video for hints]

1. What does a Hadamard gate do?
(1 mark)
2. Draw a simple quantum circuit with three qubits that returns $|111\rangle$ with certainty.
(1 mark)
3. Draw a quantum circuit with three qubits that returns $|111\rangle$ or $|101\rangle$ with 50% probability each.
(1 mark)
4. Draw a quantum circuit with three qubits that returns $|000\rangle$, $|001\rangle$, $|010\rangle$, $|011\rangle$, $|100\rangle$, $|101\rangle$, $|110\rangle$ and $|111\rangle$ with equal probability.
(2 mark)
5. Make a quantum circuit whose final Bell state is $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.
(4 marks)
6. Simulate a coin toss using qubit(s) in a quantum circuit. Why does this lead to better '[randomness](#)' than a normal/classical coin toss? How could you extend this to be a random number generator?
(4 marks)
7. Implement a solution to the Deutsch oracle problem as discussed in lectures in Qiskit (using the [Deutsch-Jozsa Algorithm](#)). Description in section 15.4.1 in (Moore and Mertens, 2011) may also be useful.
(7 marks)

For Interest Only

One can break RSA encryption 'easily' using quantum computers and [Shor's algorithm](#) in section 15.5 of (Moore and Mertens, 2011)!

References

Moore, C., Mertens, S., 2011. The Nature Of Computation. Oxford University Press.

End of Laboratory